Sparse and differentiable partial optimal transport plans through sliced approaches

Sliced optimal transport scales by projecting distributions onto random lines, where transport is solved in closed form; partial optimal transport relaxes mass conservation to compare distributions of unequal mass. Existing sliced partial methods only ever return a *distance* (or reweighted marginals), so the fact that different slices discard different mass never has to be reconciled. However, this also means they produce no transport *plan* in the original space, and averaging the per-slice plans would destroy the sparsity that makes partial OT attractive. The internship aims at a sliced partial OT that yields a sparse, differentiable plan in the original space.

§1. Organization

- **Start date**: As soon as possible (flexible).
- **Localization**: IRISA, Rennes.
- **Supervision**:
  - Romain Tavenard (Prof., Université Rennes 2, MALT Inria team, IRISA, Rennes).
  - Laetitia Chapel (Prof., Institut Agro, TAGADA research group, IRISA, Vannes).

§2. Context

Optimal transport (OT) provides a principled geometry for comparing probability distributions, but its cubic cost limits its use at scale. *Sliced* OT sidesteps this by projecting the distributions onto many random one-dimensional lines, solving the resulting transport problems in closed form via sorting, and averaging. *Partial* OT, in turn, relaxes the mass-conservation constraint so that only a fraction of the mass needs to be transported, which is especially valuable when distributions differ in total mass or contain outliers (Bonneel & Coeurjolly, 2019; Bai et al., 2023).

Two ingredients are now efficient: partial Wasserstein distances on the line admit a fast solution (Chapel & Tavenard, 2025; Angrick et al., 2026), and existing sliced partial and unbalanced variants (Bai et al., 2023; Séjourné et al., 2023) are GPU-friendly. Crucially, though, these methods are designed as *distances*: they integrate the one-dimensional partial cost over projections, and what they recover in the original space is at most reweighted marginals (the inputs with outlier mass discarded), never a coupling. Séjourné et al. even note that per-slice mass selection makes the retained marginals *inconsistent across projections*, and sidestep it by reweighting the inputs globally. A separate line lifts a *single* optimized projection's one-dimensional plan back to the original space through a differentiable bilevel formulation (Chapel et al., 2025), yielding a sparse plan (a permutation), but only in the *balanced* case.

§3. Scientific challenge

The two lines expose a gap. Averaging-based sliced partial OT never has to reconcile the different supports selected by different slices, precisely because it only computes a distance; but for the same reason it produces no plan in the original space, and naively averaging the per-slice plans would yield a *dense* coupling, destroying the sparse, interpretable matching that is the whole point of discarding outlier mass. Lifting a single projection's plan keeps that sparsity and is differentiable, but has only been done for balanced OT. The central question is therefore: **can a single-projector lift be extended to the partial setting, so that sliced partial OT returns a sparse, differentiable transport plan in the original space?** The obstacle is the mass-selection step: which mass a slice retains is a combinatorial choice, piecewise constant in the projection, so the lifted plan jumps as the projector moves.

The end goal is a sliced partial OT that returns a sparse, differentiable plan in the original space. Concretely, the candidate will:

1. formalize a partial one-dimensional plan and its lift to the original space, characterizing how the mass-selection step makes the lifted plan discontinuous in the projection direction;
2. extend the single-projector bilevel lift (Chapel et al., 2025) to the partial setting and assess what approximation guarantees survive;
3. benchmark against exact (non-sliced) partial OT and against averaging-based sliced partial baselines (Bai et al., 2023; Séjourné et al., 2023) on problems small enough to admit computable ground truth, measuring plan sparsity, and cost accuracy;
4. validate the estimator as a differentiable loss / plan in a downstream task (e.g. robust point-cloud matching or generative alignment), reporting sparsity, stability, and convergence.

Candidates should possess:

- good Python coding skills,
- knowledge of the fundamentals of optimal transport and numerical optimization,
- familiarity with the POT (Python Optimal Transport) library is a plus,
- ability to communicate in English in a scientific context.

Funding for a Ph.D. has already been secured: upon a successful internship, the candidate will be offered the opportunity to continue this work as a fully funded doctoral thesis. A successful candidate will develop or reinforce skills key to that continuation, namely:

- a practical and theoretical understanding of sliced and partial optimal transport, including transport plans and their lifts,
- the ability to turn a structural obstacle into a differentiable algorithm,
- experience benchmarking differentiable OT losses and plans for downstream training.

§4. Bibliography

Angrick, S., Conradi, J., Csikós, M., Hastrich, N., Mittal, D., Nusser, A., Onak, K., & Raghvendra, S. (2026). Computing All Optimal Partial p-Wasserstein Matchings on the Line. arXiv:2608.18875.

Bai, Y., Schmitzer, B., Thorpe, M., & Kolouri, S. (2023). *Sliced Optimal Partial Transport.* CVPR.

Bonneel, N., & Coeurjolly, D. (2019). *SPOT: Sliced Partial Optimal Transport.* ACM Transactions on Graphics.

Chapel, L., & Tavenard, R. (2025). *One for All and All for One: Efficient Computation of Partial Wasserstein Distances on the Line.* ICLR.

Chapel, L., Tavenard, R., & Vaiter, S. (2025). *Differentiable Generalized Sliced Wasserstein Plans.* NeurIPS. arXiv:2505.22049.

Nguyen, K., et al. (2025). *Sparse Partial Optimal Transport via Quadratic Regularization.* arXiv:2508.08476.

Séjourné, T., Bonet, C., Fatras, K., Nadjahi, K., & Courty, N. (2023). *Unbalanced Optimal Transport meets Sliced-Wasserstein.* arXiv:2306.07176.